

 a.k.a. “daylily”

Email i@dayli.ly

Phone 

GitHub [coclique](#)

LinkedIn 

Education

Carnegie Mellon University, *B.S. in Computer Science (GPA 4.00)*

Pittsburgh, PA · Aug 2023 – Dec 2026

- Coursework in: programming language theory, compiler design, database systems, parallel computing, distributed systems, category theory, introductory theory of computation, introductory HCI, vector calculus, statistics

Work Experience

Undisclosed financial institution, *Dev Intern*

New York, NY · May 2026 – Aug 2026

- I cannot discuss details about this internship because my work is yet to conclude

Google, *Software Engineering Intern (Travel Ads)*

Mountain View, CA · May 2025 – Aug 2025

STEP Intern (Google Learning)

New York, NY · May 2024 – Aug 2024

- (2025) Built a bidding simulation pipeline that verified the increased earning potential of ROAS bidding in Hotel Ads
- (2024) Implemented the key concept cards feature in the Google Search math solver and a related Gemini integration

MLabs Ltd., *Software Consultant (Haskell & Plutus)*

London, UK (Remote) · Oct 2021 – May 2023

- Developed an embedded DSL in Haskell for writing and visualizing [Cardano blockchain](#) transaction specifications
- Implemented unit and property testing libraries for the Plutus smart contract language

Taichi Graphics, *Intern Compiler Engineer*

Beijing, China · Nov 2021 – Mar 2023

- Proposed and implemented a type system for the [Taichi compiler](#), in place of ad-hoc validation logic
- Implemented various smaller-scale refactors and compiler passes, such as short-circuiting logical operators

Chinese Academy of Sciences, *Intern Research Engineer (PLCT Lab)*

Beijing, China · Oct 2020 – Oct 2021

- Core contributor to the open source HoTT proof assistant [Aya Prover](#), mainly on its module and type systems

Personal Projects

Cleff

<https://github.com/coclique/cleff>

- Efficient and expressive effect handlers in Haskell, 2–5x performance gain compared to mainstream libraries
- Presented at ; project gained 100+ stars on GitHub

Inkclip

<https://github.com/coclique/inkclip>

- A lightweight e-ink wearable accessory as an exercise in blending technology and self-expression
- Custom STM32-based PCB, Rust ([embassy](#)) firmware, and [Svelte](#)-based web interface using [WebHID](#) communication

Lilynet (AS401736)

<https://daylily.network>

- Educational dual-stack (IPv4+v6) network participating in global internet routing, active voting member of [ARIN](#)
- 3 virtual PoPs providing services including authoritative DNS, Tor relays, IPv6 tunneling, and anycast routing

Miscellaneous

- **Specializations:** functional programming; compilers and type systems; full-stack web development; among others
- **Programming languages:** Haskell, TypeScript, HTML/CSS, OCaml, Kotlin, Java, C/C++, Rust, Python, LaTeX, etc.
- **Natural languages:** bilingual in English and Mandarin